Positioning and Mounting

The loads on the roof are to be considered, if turbines should be mounted on already existing houses. The main problem is the turbulent area on roof tops (see Figure 8).

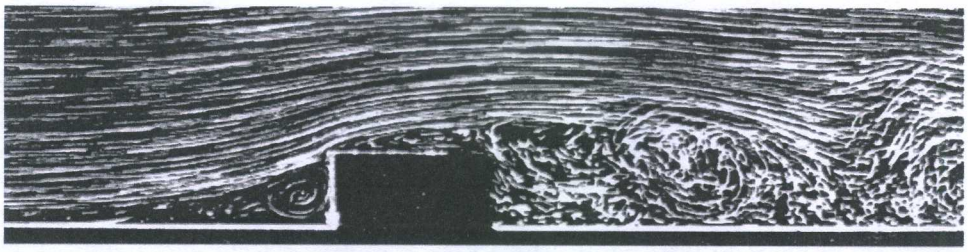

Figure 8: Flow visualized around a squared building [xii]

With the turbulence come big changes in loads and inefficient power yield. Therefore high towers are necessary to escape the most turbulent area. Linked to a high tower are high moments induced to the roof structure. That requires a reasonable strong fixing point and a structural analysis of the roof. Prevention or reduction of vibration is achieved by rubber mounts, applied by e. g. Murray O` Laoire Associates for the turbines on Green Building in Dublin (see Figure 9, left) as well as by Prowins`s Provane[xiii].

Figure 9: left: Green House in Dublin [xiv]; middle: Swift turbine mounting pole [xv]; right: ARUP`s VAWTEX [xvi]

Proven declares, that vibrations are no problem, if a strong fixing point into a concrete reinforced beam or similar is available, while they use a soft mount system between tower and roof structure to supply additional damping. To come back to the flow field around a building, see Figure 10, besides the high turbulent area on the flat roof top, the walls are subjected to high pressure and low pressure sides respectively, depending on the wind direction. With these pressure differences accelerated and decelerated wind fields are present. Where the suction areas are, an acceleration of the flow will take place.

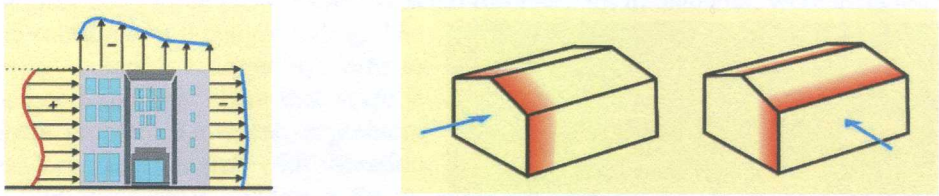

Figure 10: Left: pressure distribution around building; right: wind direction depending suction areas shown by the red colour [xvii]

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Risø-R-Report

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